

ET65H11XX - High Input Very-Low IQ 500mA LDO

General Description

ET65H11XX are the high input very low I_Q 500mA LDO with high power supply rejection ratio (PSRR), is designed specifically for portable battery-powered applications which require ultra-low quiescent current. The very-low consumption of type $1\mu A$ ensures long battery life and dynamic transient boost feature improves device transient response for wireless communication applications.

The ET65H11XX family are available in standard fixed output voltages of 1.8V (ET65H1118), 2.5V (ET65H1125), 2.8V (ET65H1128), 3.0V (ET65H1130), 3.3V (ET65H1133), 3.6V (ET65H1136), 4.0V (ET65H1140), 5.0V (ET65H1150), and custom voltage options (50mV step options).

ET65H11XX is offered SOT89-3, SOT23-5, SOT23-3 packages.

Features

- Wide Input Voltage Range: 1.8V to 12V
- Up to 500mA Load Current
- Standard Fixed Output Voltage Options: 1.8V, 2.5V, 2.8V, 3.0V, 3.3V, 3.6V, 4.0V, and 5.0V
- Other Output Voltage Options Available on Request
- Very Low I_Q : $1\mu A$
- Low Dropout: 730mV @ 500mA / $V_{OUT}=3.3V$
- Very High PSRR: 80dB at 1KHz
- Excellent Load/Line Transient Response
- ESD Protection Complies with JESD22 Standard
 - HBM: $\pm 4000V$ Pass (JEDEC JS-001)
 - CDM: $\pm 1500V$ Pass (JEDEC JS-002)
- Latch-up Performance Exceeds $\pm 200mA$ per JEDEC JESD78F
- Part No. and Package Information

Part No.	Package	Packing Option	MSL
ET65H11XXB	SOT89-3	Tape and Reel, 1K/ Reel	3
ET65H11XXS	SOT23-3	Tape and Reel, 3K/ Reel	3
ET65H11XX	SOT23-5	Tape and Reel, 3K/ Reel	3

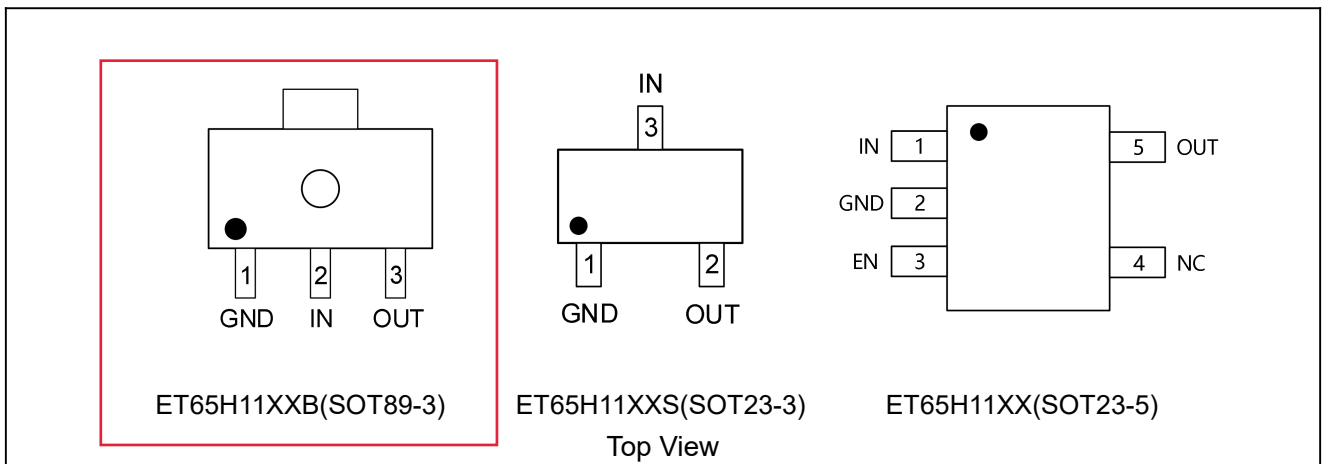
Device information

ET 65H11 XX X

<u>XX</u> Output Voltage		<u>X</u> Package	
XX	Output X.X V	B	SOT89-3
		S	SOT23-3
		/	SOT23-5 (Default)

ET65H11XX

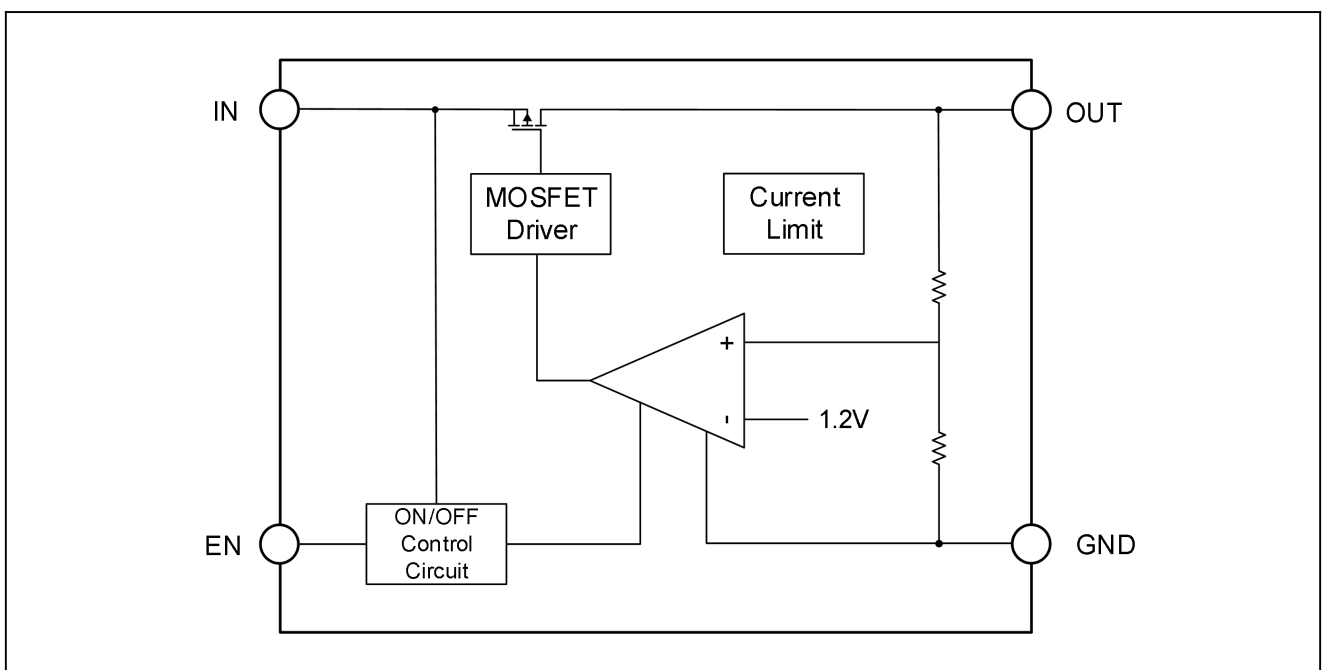
Pin Configuration



Pin Function

Pin No.			Pin Name	Pin Function
SOT89-3	SOT23-3	SOT23-5		
XXB	XXS	XX		
1	1	2	GND	Ground.
2	3	1	IN	Supply Input Pin.
3	2	5	OUT	Output Pin.
		3	EN	Enable Control Input, Active High.
		4	NC	No Connection.

Block Diagram



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Functional Description

Input Capacitor

A 1.0 μ F ~ 10 μ F ceramic capacitor is recommended to connect between IN and GND pins to decouple input power supply glitch and noise. The amount of the capacitance may be increased without limit. This input capacitor must be located as close as possible to the device to assure input stability and less noise. For PCB layout, a wide copper trace is required for both IN and GND.

Output Capacitor

An output capacitor is required for the stability of the LDO. The recommended output capacitance is from 1.0 μ F ~ 10 μ F, Equivalent Series Resistance (ESR) is from 5m Ω to 100m Ω , and temperature characteristics are X7R or X5R. Higher capacitance values help to improve load/line transient response. The output capacitance may be increased to keep low undershoot/overshoot. Place output capacitor as close as possible to OUT and GND pins.

Enable

The ET65H11XX delivers the output power when it is set to enable state. When it works in disable state, there is no output power and the operation quiescent current is almost zero. The enable pin (EN) is active high.

Dropout Voltage

The ET65H11XX uses a PMOS pass transistor to achieve low dropout. When ($V_{IN} - V_{OUT}$) is less than the dropout voltage (V_{DROP}), the PMOS pass device is in the linear region of operation and the input-to-output resistance is the $R_{DS(ON)}$ of the PMOS pass element. V_{DROP} scales approximately with output current because the PMOS device behaves like a resistor in dropout mode. As with any linear regulator, PSRR and transient response degrade as ($V_{IN} - V_{OUT}$) approaches dropout operation.

Thermal Shutdown

Thermal shutdown protection disables the output when the junction temperature rises to approximately 150°C. Disabling the device eliminates the power dissipated by the device, allowing the device to cool. When the junction temperature cools to approximately 125°C, the output circuitry is again enabled. Depending on power dissipation, thermal resistance, and ambient temperature, the thermal protection circuit may cycle on and off. This cycling limits regulator dissipation, protecting the LDO from damage as a result of overheating. Activating the thermal shutdown feature usually indicates excessive power dissipation as a result of the product of the ($V_{IN} - V_{OUT}$) voltage and the load current. For reliable operation, limit junction temperature to 125°C maximum.

Thermal Considerations

For continuous operation, do not exceed absolute maximum junction temperature. The maximum power dissipation depends on the thermal resistance of the IC package, PCB layout, rate of surrounding airflow, and difference between junction and ambient temperature. The maximum power dissipation can be calculated by the following formula:

$$P_{D(MAX)} = (T_{J(MAX)} - T_A) / \theta_{JA}$$

where $T_{J(MAX)}$ is the maximum junction temperature, T_A is the ambient temperature, and θ_{JA} is the junction to ambient thermal resistance. The maximum power dissipation depends on the operating ambient temperature for fixed $T_{J(MAX)}$ and thermal resistance, θ_{JA} .

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Current-Limit Protection

The ET65H11XX provides current limit function to prevent the device from damages during over-load or shorted-circuit condition. This current is detected by an internal sensing transistor.

Absolute Maximum Ratings

Symbol	Rating	Value	Unit
V_{IN}	Input Voltage	-0.3~16	V
V_{OUT}	Output Voltage	-0.3~7	V
V_{EN}	Enable Input Voltage	-0.3~16	V
$T_{J(MAX)}$	Maximum Junction Temperature	150	°C
T_{STG}	Storage Temperature	-65~150	°C
V_{ESD}	Human Body Model HBM (JEDEC JS-001)	±4000	V
	Charged Device Model (JEDEC JS-002)	±1500	V
I_{LU}	Latch Up Current Maximum Rating (JEDEC JESD78F)	±200	mA

Thermal Characteristics

Symbol	Package	Ratings	Value	Unit
$R_{\theta JA}$	SOT89-3	Thermal Characteristics, Thermal Resistance, Junction-to-Air	135	°C/W
	SOT23-5		250	
	SOT23-3		360	
P_D	SOT89-3	Power Dissipation@25°C	920	mW
	SOT23-5	PCB Board Dimension:	500	
	SOT23-3	50mm x 50mm (2layer) Copper :1oz	420	

Recommended Operating Conditions

Symbol	Item	Rating	Unit
V_{IN}	Input Voltage	1.8 to 12	V
I_{OUT}	Output Current	0 to 500	mA
T_J	Operating Junction Temperature	-40 to 125	°C
C_{IN}	Effective Input Ceramic Capacitor Value	1~10	μF
C_{OUT}	Effective Output Ceramic Capacitor Value	1~10	μF
ESR	Input and Output Capacitor Equivalent Series Resistance (ESR)	5 to 100	mΩ

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Electrical Characteristics

($V_{IN} = V_{OUT} + 2V$; $I_{OUT} = 1mA$, $C_{IN} = C_{OUT} = 1.0\mu F$, unless otherwise noted. Typical values are at $T_A = 25^\circ C$.)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
$V_{IN}^{(1)}$	Operating Input Voltage		1.8		12	V
V_{OUT}	Output Voltage	$I_{OUT} = 1mA$, $25^\circ C$	-2%		+2%	V
I_{Q_ON}	Quiescent Current	$I_{OUT} = 0mA$, $25^\circ C$		1.0	2.0	μA
I_{Q_OFF}	Standby Current	$V_{EN} = 0V$, $25^\circ C$		0.2	0.5	μA
Reg_{Line}	Line Regulation	$V_{IN} = V_{OUT} + 1V$ to $12V$, $I_{OUT} = 10mA$		0.02	0.1	%/V
Reg_{LOAD}	Load Regulation	$1mA \leq I_{OUT} \leq 500mA$, $V_{IN} = V_{OUT} + 1V$		15	45	mV
$V_{DROP}^{(2)}$	Dropout Voltage	$V_{OUT} = 3.0V$, $I_{OUT} = 100mA$		120	200	mV
		$V_{OUT} = 3.0V$, $I_{OUT} = 300mA$		380	580	
		$V_{OUT} = 3.0V$, $I_{OUT} = 500mA$		750	980	
		$V_{OUT} = 3.3V$, $I_{OUT} = 100mA$		115	190	
		$V_{OUT} = 3.3V$, $I_{OUT} = 300mA$		375	560	
		$V_{OUT} = 3.3V$, $I_{OUT} = 500mA$		730	960	
		$V_{OUT} = 5.0V$, $I_{OUT} = 100mA$		110	180	
		$V_{OUT} = 5.0V$, $I_{OUT} = 300mA$		370	550	
		$V_{OUT} = 5.0V$, $I_{OUT} = 500mA$		720	950	
I_{LMT}	Current Limit	$V_{IN} = V_{OUT} + 2V$		1050	1450	mA
I_{SHORT}	Short Current Limit	$V_{IN} = V_{OUT} + 2V$, $V_{OUT} = 0V$		150	240	mA
V_{ENH}	EN Pin Threshold Voltage	EN Input Voltage Rising	1.4			V
V_{ENL}	EN Pin Threshold Voltage	EN Input Voltage Falling			0.4	V
I_{EN}	EN Pin Current	$V_{EN} = 0 \sim 12V$		0.1	1	μA
$PSRR^{(3)}$	Power Supply Rejection Ratio	$f = 1 kHz$, $V_{IN} = V_{OUT} + 2V$ $I_{OUT} = 20mA$		80		dB
$e_N^{(3)}$	Output Noise Voltage	$V_{IN} = V_{OUT} + 2V$, $I_{OUT} = 1mA$, $f = 10Hz$ to $100KHz$, $V_{OUT} = 3.3V$, $C_{OUT} = 1\mu F$		90		μV_{rms}
t_{ON}	Turn-On Time	From assertion of V_{EN} to $V_{OUT} = 90\%V_{OUT(NOM)}$		1		ms
$T_{TSD}^{(3)}$	Thermal Shutdown Temperature	Temperature Increasing from $T_A = +25^\circ C$		155		$^\circ C$
$T_{HYS}^{(3)}$	Thermal Shutdown Hysteresis	Temperature Falling from T_{TSD}		25		$^\circ C$

Note1: Here V_{IN} means internal circuit can work normal. If $V_{IN} < V_{OUT}$, Output voltage follows V_{IN} ($I_{OUT} = 1mA$), circuit is safety.

Note2: V_{DROP} FT test method: test the V_{OUT} voltage at $V_{OUT} + V_{DROPMAX}$ with 300mA output current.

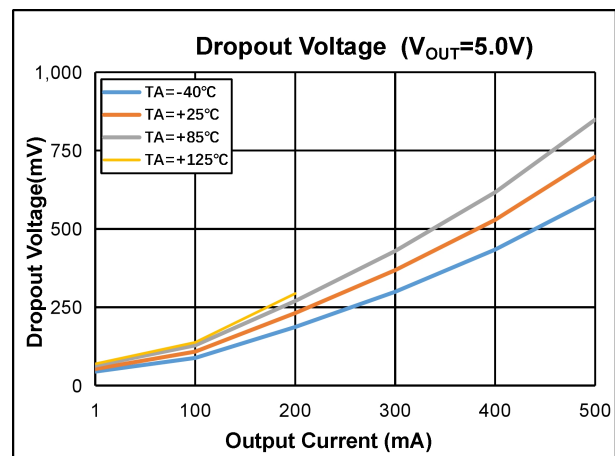
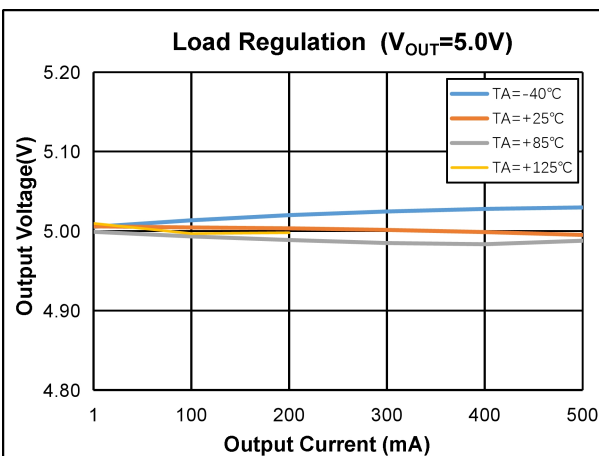
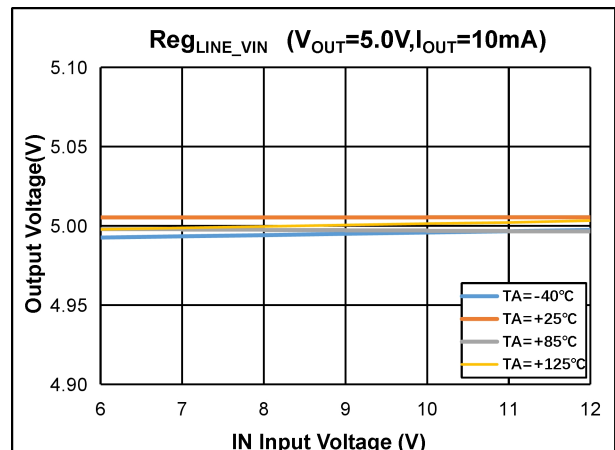
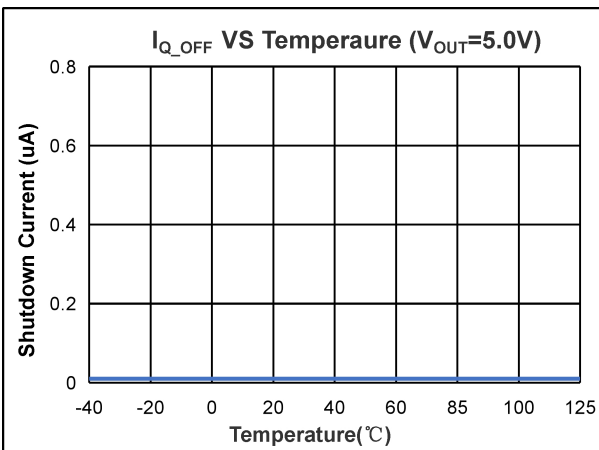
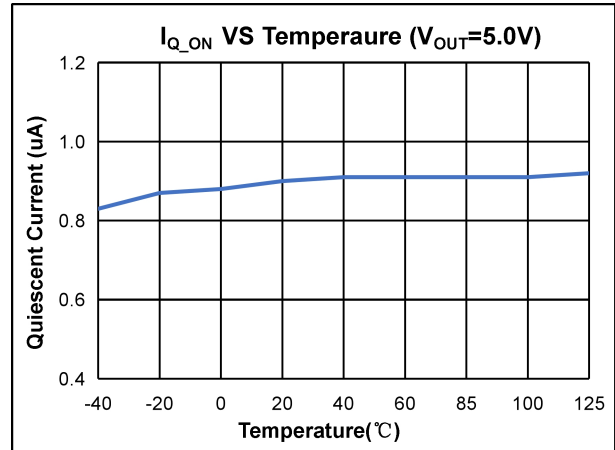
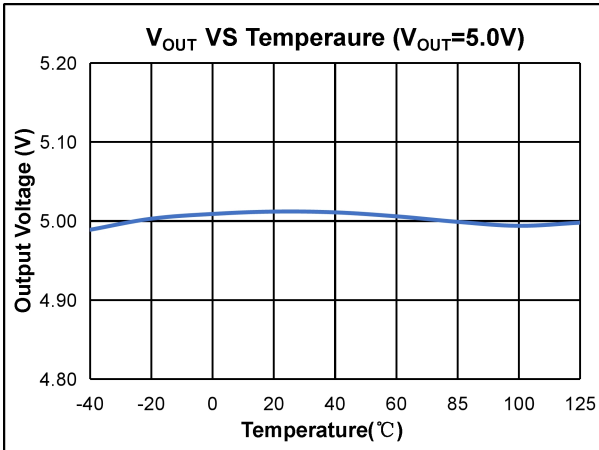
Note3: Guaranteed by design and characterization. Not a FT item.

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Typical Characteristics

VOLTAGE VERSION 5.0V

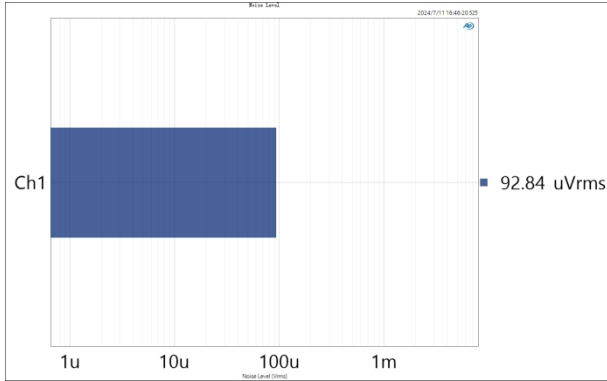
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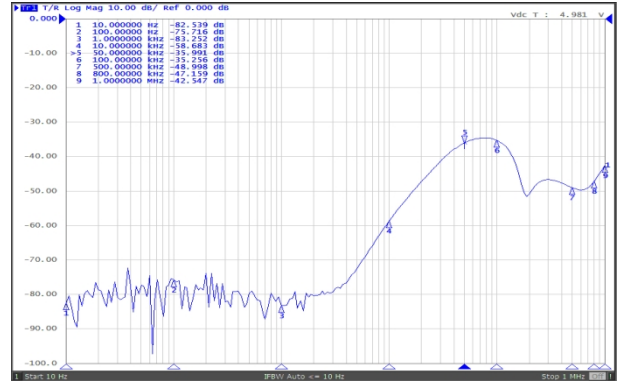
ET65H11XX

VOLTAGE VERSION 5.0 V(Continued)

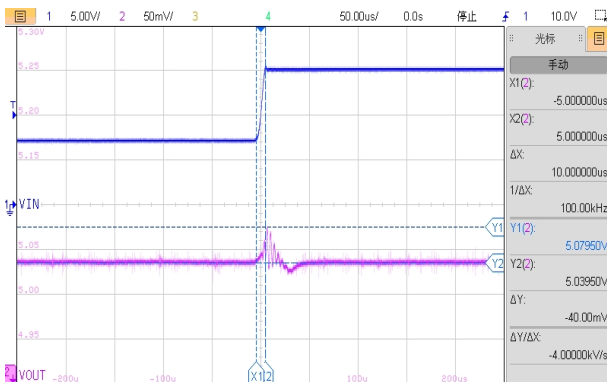
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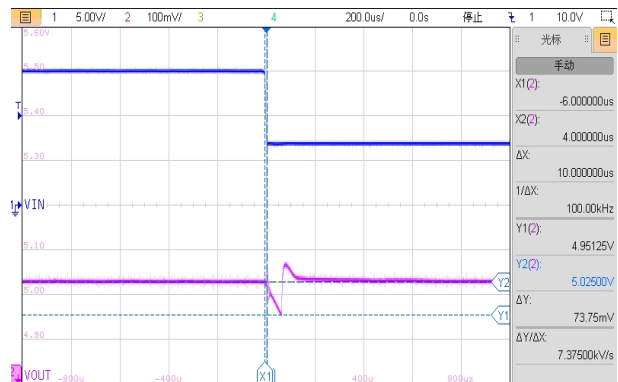
Noise($V_{IN} = 7.0V$ $I_{OUT} = 20mA$)



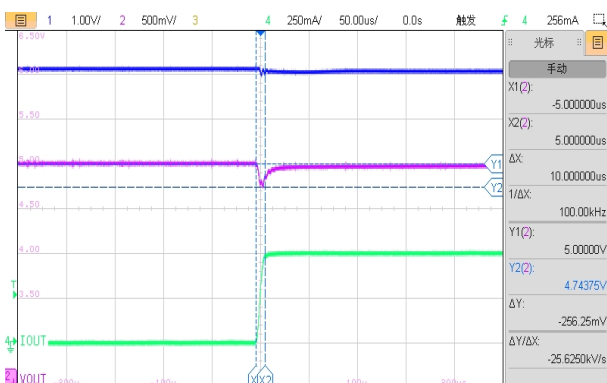
PSRR($V_{IN} = 7.0V$ $I_{OUT} = 20mA$)



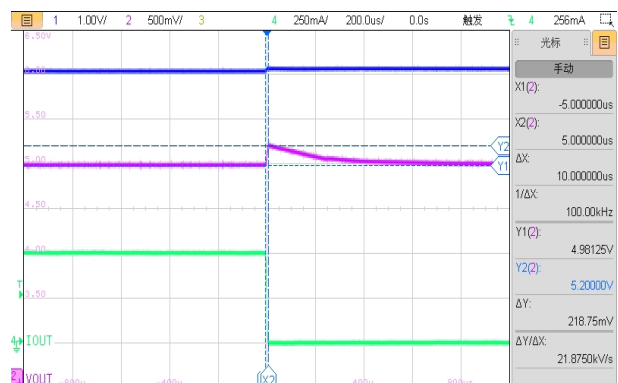
Line Transient($V_{IN} = 7.0 \sim 15V$ $I_{OUT} = 1mA$)



Line Transient($V_{IN} = 15 \sim 7.0V$ $I_{OUT} = 1mA$)



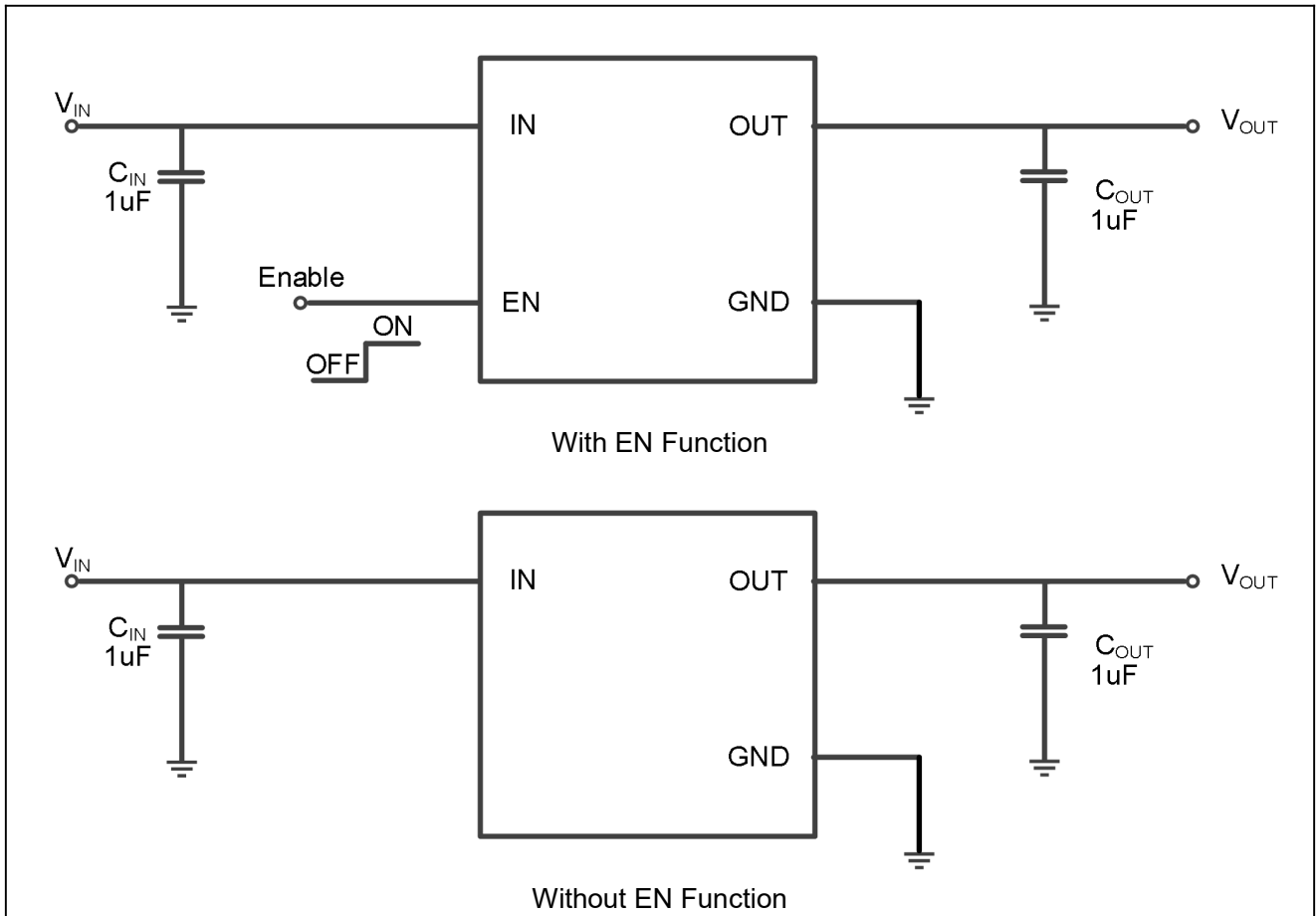
Load Transient($V_{IN} = 7.0V$ $I_{OUT} = 1 \sim 500mA$)



Load Transient($V_{IN} = 7.0V$ $I_{OUT} = 500 \sim 1mA$)

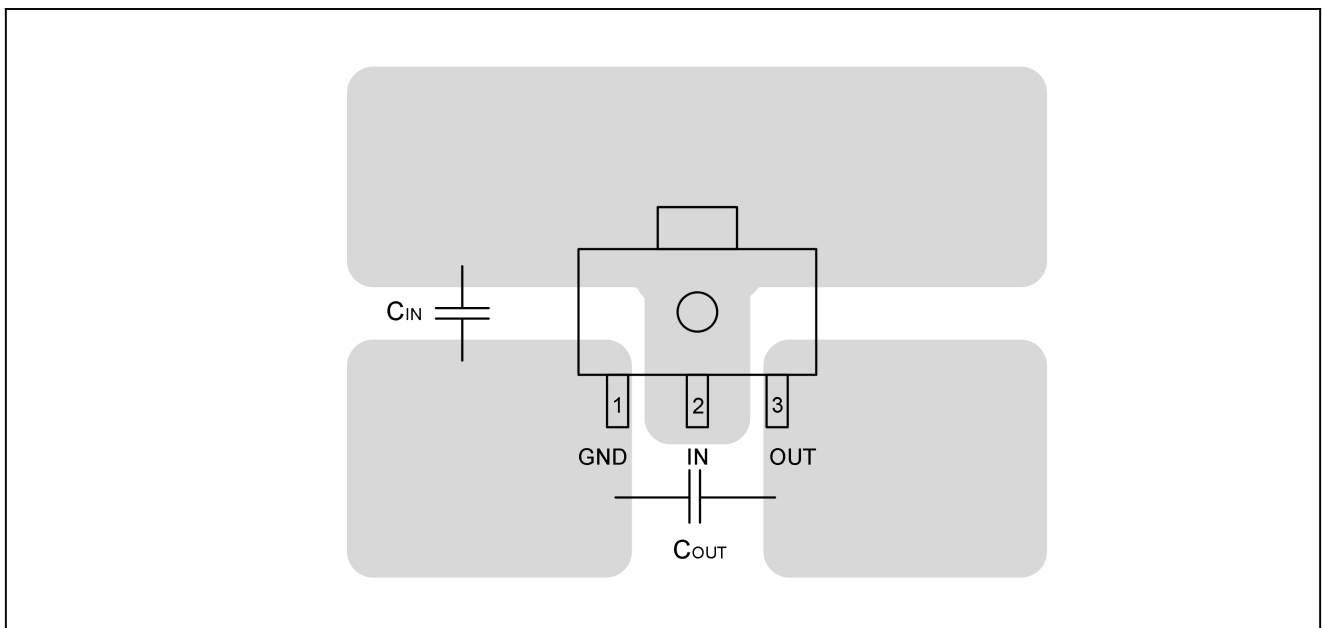
ET65H11XX

Application Circuits



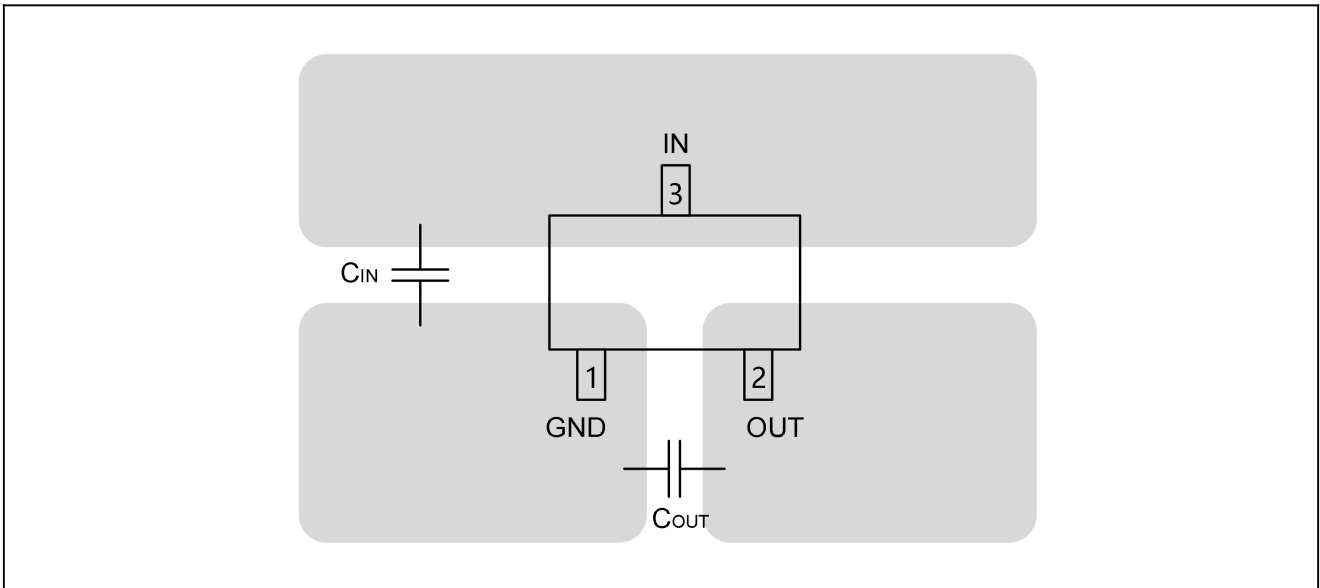
PCB Layout Guide

SOT89-3

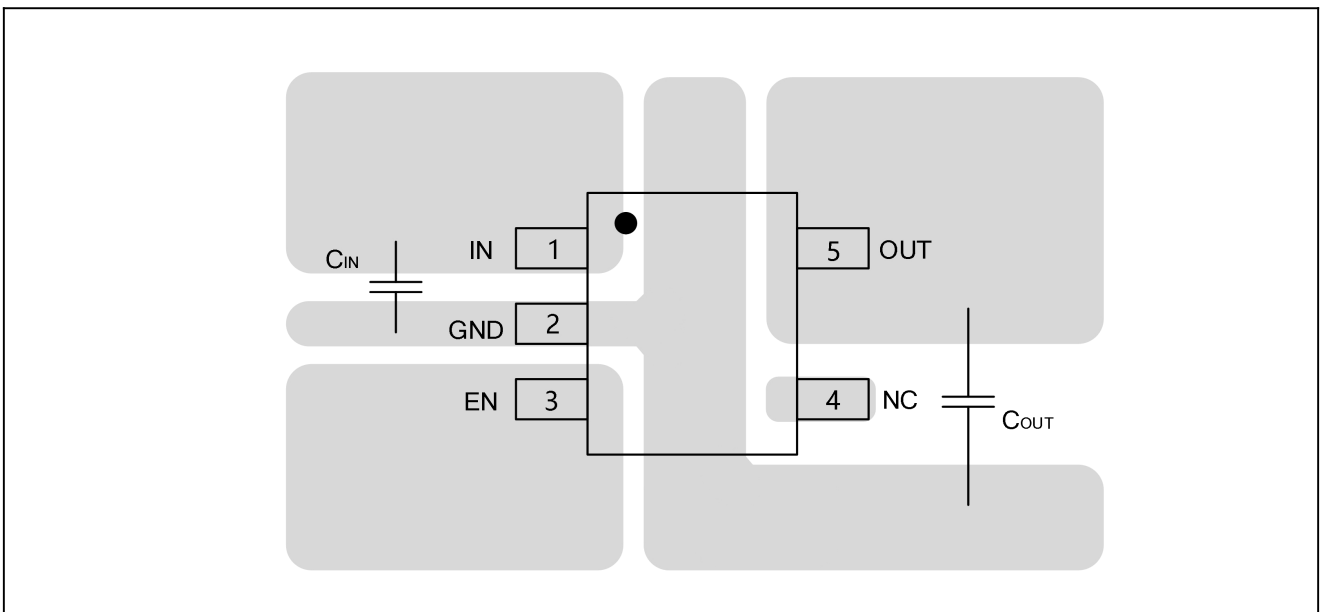


ET65H11XX

SOT23-3



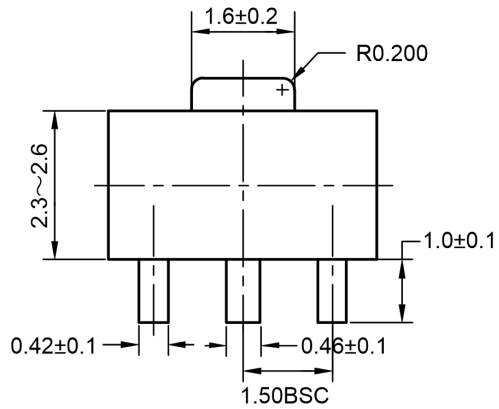
SOT23-5



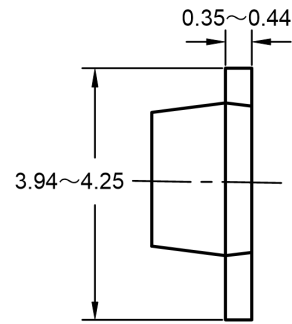
ET65H11XX

Package Dimension

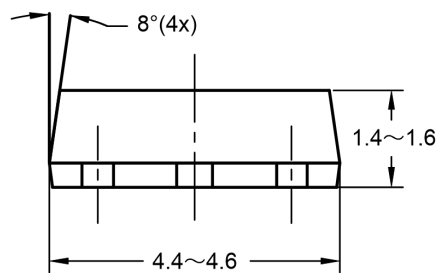
SOT89-3



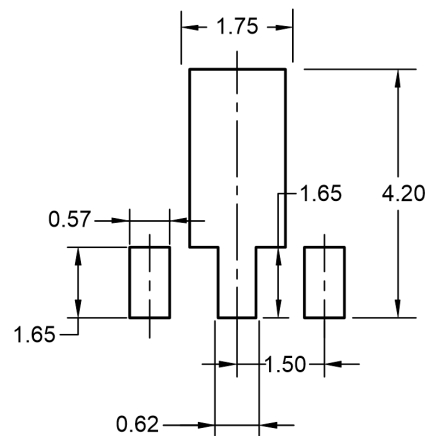
Top View



Side View



Side View

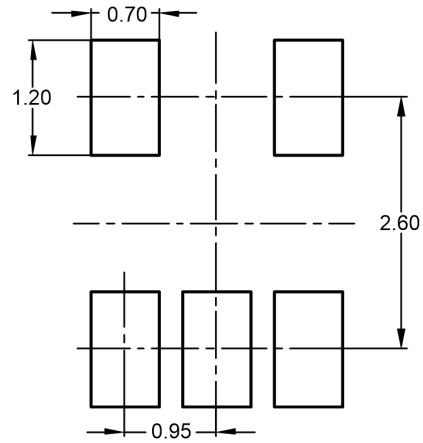
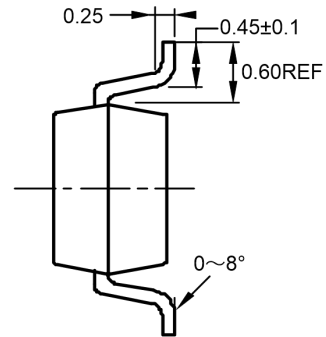
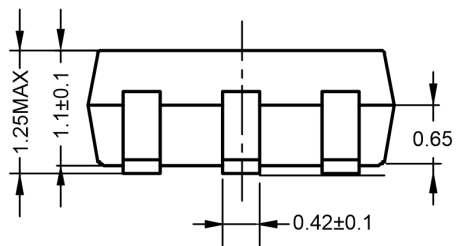
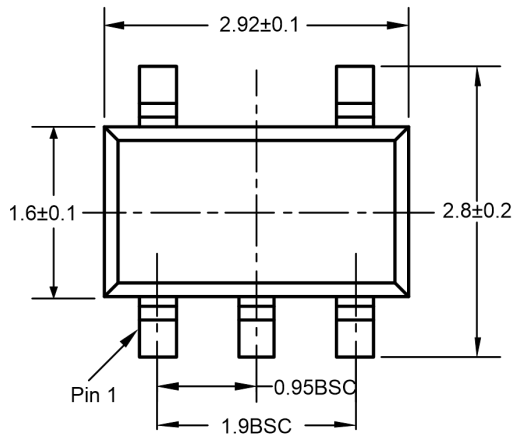


Recommended Land Pattern

Unit: mm

ET65H11XX

SOT23-5

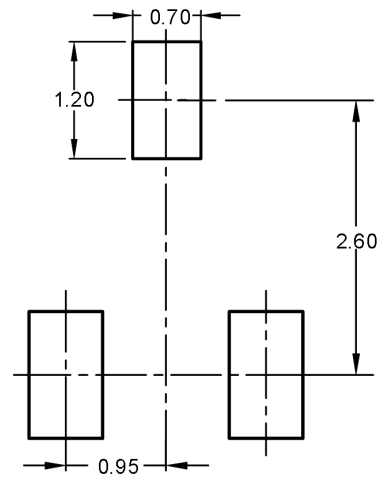
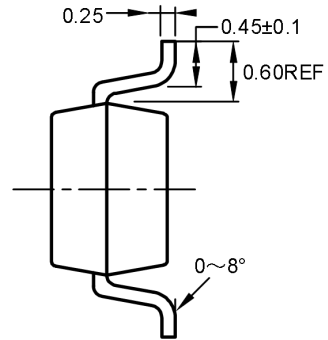
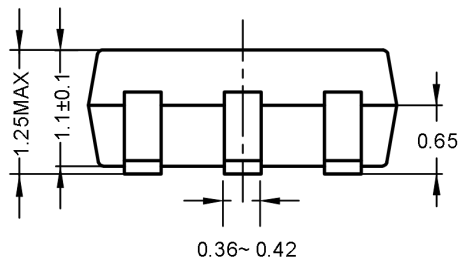
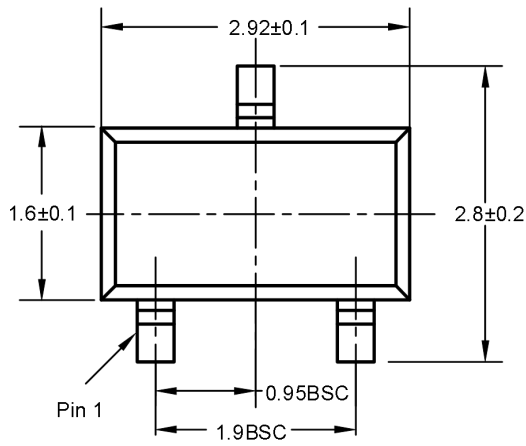


Recommended Land Pattern

Unit: mm

ET65H11XX

SOT23-3

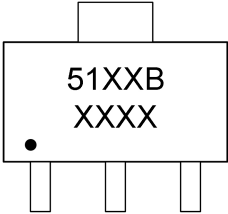
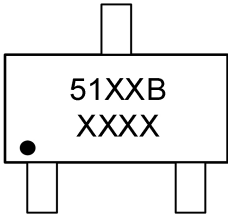
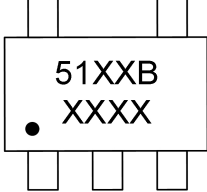


Recommended Land Pattern

Unit: mm

ET65H11XX

Marking Information

		
ET65H11XXB	ET65H11XXS	ET65H11XX
51 & B = Part Number	51 & B = Part Number	51 & B = Part Number
XX = V _{OUT} Version	XX = V _{OUT} Version	XX = V _{OUT} Version
XXXX = Tracking Number	XXXX = Tracking Number	XXXX = Tracking Number

Revision History and Checking Table

Version	Date	Revision Item	Modifier	Function & Spec Checking	Package & Tape Checking
0.0	2024-02-21	Preliminary Version	Pengjj	Liuxm	Liujy
1.0	2024-08-03	Official Version	Tugz	Liuxm	Liujy
1.1	2024-12-19	Update Vdrop	Tugz	Liuxm	Liujy
1.1.a	2026-03-16	Update Format	Yuyf	Yangxx	Liujy
1.2	2026-04-15	Add Marking	Xut	Yangxx	Liujy